

Airfoil analysis using XFLR5

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Abstract

This laboratory investigates the aerodynamic performance of four laminar-flow airfoils from the NACA 6-series—NACA 63018, NACA 63012, NACA 63009, and NACA 6A006—using the software XFLR5. The analysis involves both the lift coefficient variation with the angle of attack, $C_l(\alpha)$, and the surface pressure coefficient distributions, $C_p(x/c)$, for selected angles of attack. The objective is to compare the computational results from XFLR5 with the experimental data provided in NACA Technical Note 2502, highlighting the similarities and discrepancies between theoretical predictions and wind tunnel measurements. The differences are discussed in terms of laminar flow extent, transition modeling, boundary layer behavior, and Reynolds number sensitivity. Through this comparison, the study reinforces understanding of airfoil performance prediction and the limitations of panel methods and simplified boundary layer models in low-speed aerodynamic analysis.

1 Introduction

Accurate prediction of aerodynamic characteristics is crucial in the design of airfoils for efficient flight. The NACA 6-series airfoils were developed to maximize laminar flow, thereby reducing skin-friction drag and improving performance at moderate lift coefficients. In this experiment, XFLR5—a computational tool based on potential flow and a 1D boundary layer model—is used to replicate and compare the lift and pressure characteristics of selected NACA 6-series profiles.

The main goal of this experiment is to analyze the selected NACA airfoils using XFLR5, obtaining both the $C_l(\alpha)$ and $C_p(x/c)$ curves under the same conditions reported in NACA TN-2502. These computational results are then compared with the experimental data to highlight and discuss any differences between the numerical predictions and the measured aerodynamic behavior.

Such deviations are expected due to simplifications in the numerical model, such as the absence of fully resolved viscous effects, fixed transition points, and simplified turbulence representation. The analysis provides insight into the capabilities and limitations of XFLR5 for preliminary aerodynamic design.

2 Methodology

2.1 Software Setup

XFLR5 couples a panel method with a one-dimensional boundary layer model. The steps followed are those described in the XFLR5 Guidelines (Section 3.2), summarized as:

1. Define or load the NACA 6-series airfoils.
2. Set up Type 1 analysis at $Re = 5.800.000$ and $M = 0.17$.
3. Define the range of angles of attack.
4. Run the analysis and export the resulting polars and pressure distributions.

2.2 Airfoils and Test Conditions

The analyzed airfoils and corresponding experimental comparison angles are:

- NACA 633018: $\alpha = 6^\circ$ and $\alpha = 16^\circ$
- NACA 631012: $\alpha = 7.8^\circ$ and $\alpha = 13.8^\circ$
- NACA 63009: $\alpha = 6^\circ$ and $\alpha = 9^\circ$
- NACA 63A006: $\alpha = 4.5^\circ$ and $\alpha = 8^\circ$

3 Results and Discussion

3.1 Pressure Distribution Analysis

The pressure coefficient distributions $C_p(x/c)$ obtained from XFLR5 for the four NACA 6-series airfoils illustrate the evolution of surface pressure with angle of attack and the influence of thickness on aerodynamic performance. The analyzed airfoils include the NACA 63A006, 63 009, 63 1012, and 63 3018, each evaluated at two distinct angles of attack.

NACA 63-006

At $\alpha = 4.5^\circ$, the pressure coefficient reaches a C_p drop of approximately $C_p \approx -8$ near the leading edge, followed by a smooth recovery toward the trailing edge. Increasing the angle to $\alpha = 7.25^\circ$ deepens this peak to nearly $C_p \approx -15$, reflecting a stronger suction and thus a higher lift coefficient. The smooth recovery indicates attached flow with no evident separation, meaning the airfoil remains in the linear lift regime. Owing to its small thickness (6% of chord), this airfoil generates lift efficiently at low drag but is expected to stall abruptly beyond moderate angles, we can assume due to the problems with converge in the program that somewhere around 7.25 and 8 degrees generates an abrupt change or instability in the boundary condition and therefore the study was done with 7.25.

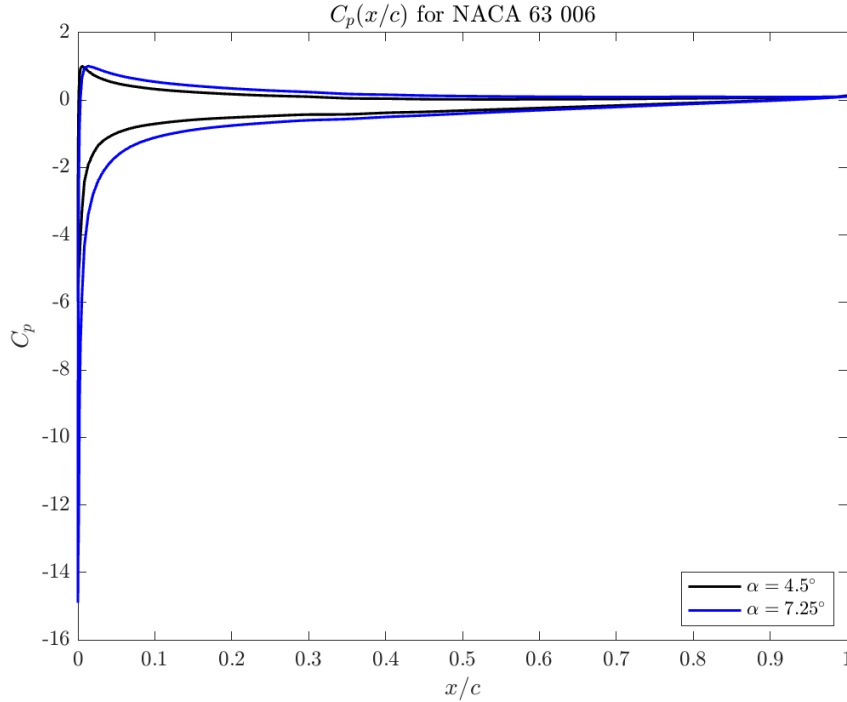


Figure 1: NACA 63A006 at $\alpha = 4.5^\circ$ and 8° .

NACA 63-009

For $\alpha = 6^\circ$, the pressure drop occurs around $C_p \approx -10$, while at $\alpha = 9^\circ$ it deepens slightly to about -12 . The overall distribution remains smooth, showing an attached boundary layer and steady lift increase. The 9% thickness moderates the pressure drop, reducing adverse pressure gradients and enhancing stall resistance compared to the thinner 63-006 profile. The recovery region remains gradual and continuous, indicating sustained laminar behavior over most of the chord.

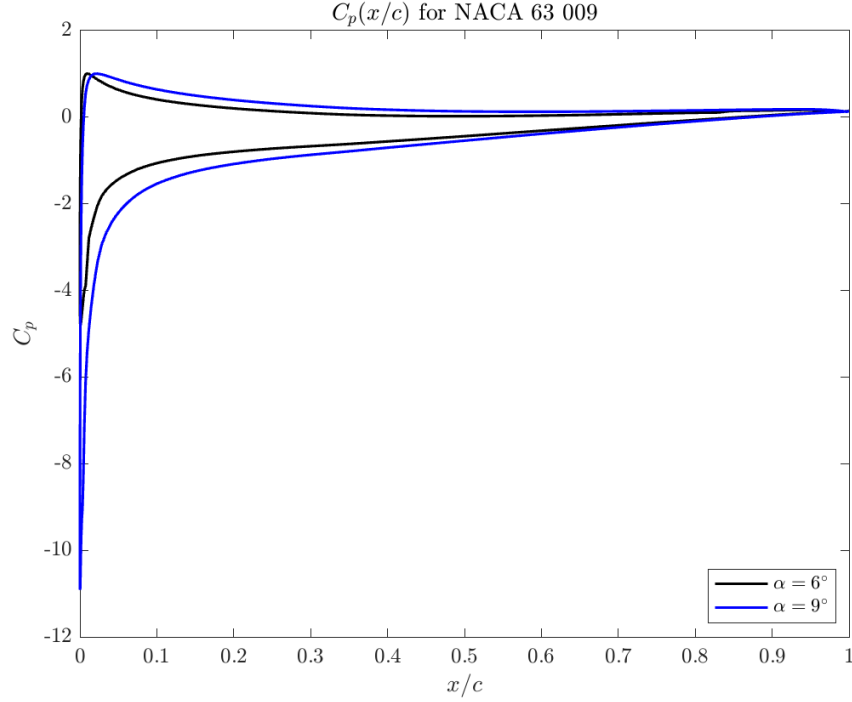


Figure 2: NACA 63009 at $\alpha = 6^\circ$ and 9° .

NACA 63-1012

At $\alpha = 7.8^\circ$, the upper-surface pressure minimum approaches $C_p \approx -10$ with a well-defined laminar region up to $x/c \approx 0.3$. Increasing the angle to $\alpha = 13.8^\circ$ deepens the minimum pressure to nearly $C_p \approx -15$, while the recovery flattens considerably, indicating that the boundary layer begins to separate near mid-chord. The lower surface pressure remains nearly constant, so lift increase slows down at higher angles. This behavior suggests that the airfoil is approaching stall, which aligns with experimental findings reported in NACA Technical Note 2502.

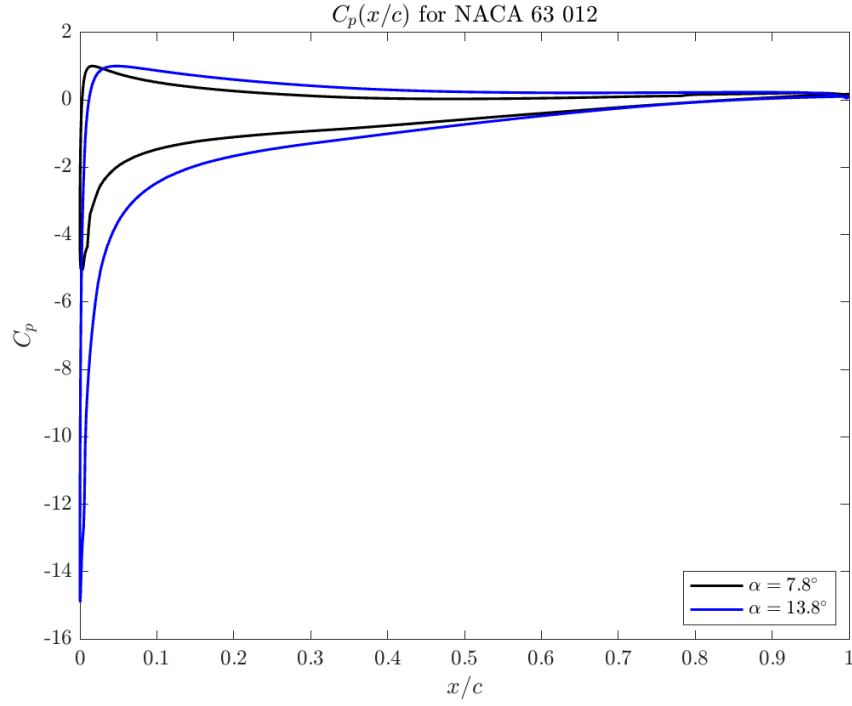


Figure 3: NACA 631012 at $\alpha = 7.8^\circ$ and 13.8° .

NACA 63-3018

For $\alpha = 6^\circ$, the suction peak is weaker, around $C_p \approx -4$, consistent with the large thickness and mild camber of this section. At $\alpha = 16^\circ$, the upper-surface pressure drops to about -8 , but the recovery region becomes distorted and flattened near the trailing edge, suggesting the onset of partial separation. The thicker geometry (18% of chord) produces a longer region of nearly constant pressure, typical of turbulent reattachment or transition effects. Consequently, while the airfoil can achieve high lift at large angles, it also experiences earlier separation than predicted by the potential flow model.

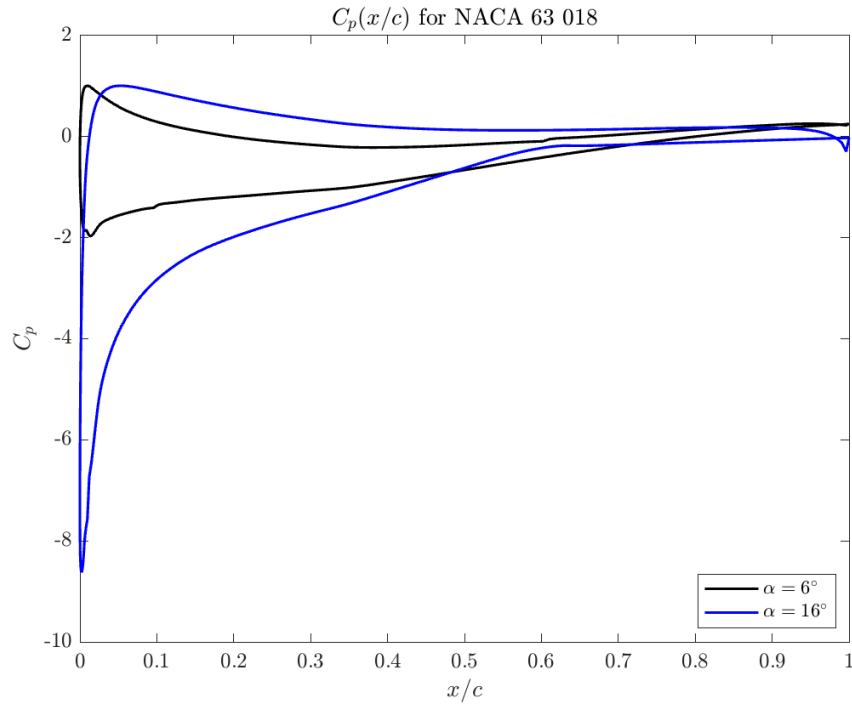


Figure 4: NACA 633018 at $\alpha = 6^\circ$ and 16° .

3.2 Comparative Trends Among Airfoils

Across the analyzed airfoils, some common tendencies are observed. When increasing the thickness, the leading-edge suction peaks but delays stall, producing smoother lift degradation at high angles. In the thinnest airfoils (63A006 and 63 009) exhibit sharp suction peaks and efficient lift generation in the pre-stall region, at the expense of abrupt stall behavior. At the thinnest one XFLR5 didn't converge further than 7,25 degrees so our analysis of the results may be limited. Thicker sections (631012 and 633018) display more gradual pressure recovery and improved flow attachment at moderate angles, consistent with more forgiving stall characteristics. The XFLR5 predictions capture these laminar trends effectively, though the software tends to underpredict flow separation and transition effects due to its simplified one-dimensional boundary layer model.

Overall, the pressure distributions confirm the expected aerodynamic behavior of the NACA 6-series airfoils: thin sections favor low-drag, high-lift laminar flow at moderate angles, while thicker sections provide higher maximum lift coefficients and smoother stall characteristics.

3.3 Lift Curve Comparison

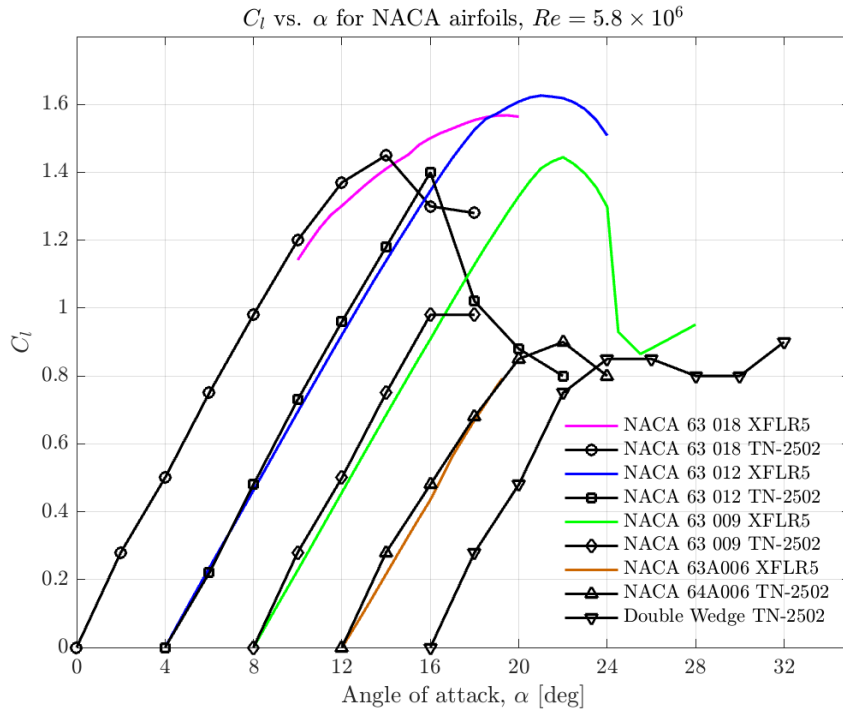


Figure 5: Lift coefficient C_l vs. angle of attack α for all four NACA airfoils.

The comparison between the computational results obtained with XFLR5 and the TN-2502 data shows a good general match in both lift and pressure behavior. For all four NACA 6-series airfoils, the general shape of the $C_l(\alpha)$ curves is consistent: a linear increase in lift with angle of attack up to the stall region, with thicker airfoils achieving higher maximum lift and smoother stall characteristics. However, the XFLR5 results tend to predict slightly higher lift coefficients and a more gradual stall compared to the experimental measurements, in which the stall occurs earlier and more abruptly—particularly for the thinner sections (NACA 63A006 and 63009). Despite these differences, the numerical results correctly capture the relative trends among the airfoils, showing how increased thickness delays stall and moderates pressure gradients.

The $C_p(x/c)$ distributions exhibit a similar level of agreement. In both XFLR5 and the TN-2502 data, increasing the angle of attack deepens the suction peak near the leading edge and flattens the pressure recovery region toward the trailing edge. Nevertheless, the experimental pressure distributions generally show weaker suction peaks and evidence of early flow separation, especially at higher angles of attack. XFLR5, in contrast, produces smoother distributions with stronger suction and delayed

separation, reflecting its assumption of attached, laminar flow. These differences are most pronounced for the thinner airfoils, where laminar separation bubbles and transition effects are more significant in the experimental data.

The differences between the experimental and numerical results mainly come from the simplifications used in XFLR5. Since the program is based on potential flow with a simplified boundary-layer model, it does not fully capture three-dimensional separation, flow transition, or the influence of surface roughness and tunnel effects seen in the experiments. The way the analysis is set up also affects the results, the number of panels used to define the airfoil or the chosen transition settings can influence how the lift and pressure distributions are predicted. On the other hand, the experimental data can also include uncertainties due to small alignment errors, surface imperfections, or measurement variations, which contribute to the differences between both datasets.

Overall, the comparison confirms that XFLR5 provides a reliable qualitative prediction of the aerodynamic behavior of laminar-flow airfoils, capturing the influence of airfoil thickness and angle of attack on lift and pressure distribution. However, its limitations in modeling viscous and transitional effects lead to optimistic predictions near stall.

4 Discussion

The results from XFLR5 demonstrate that the software provides a reasonable approximation of the aerodynamic behavior of NACA 6-series airfoils, capturing the general lift trends and the evolution of surface pressure with angle of attack. Thinner profiles such as the NACA 63A006 and 63009 exhibited strong leading-edge pressure gradients and high lift efficiency at moderate angles, whereas thicker sections like the 631012 and 633018 showed milder gradients and more gradual stall characteristics. The discrepancies between numerical and experimental data primarily stem from XFLR5's simplified potential-flow model, which assumes attached laminar flow and neglects three-dimensional separation and transition effects. As a result, the computational predictions tend to overestimate lift near stall. Despite these limitations, the analysis highlights the expected influence of airfoil thickness on pressure recovery and stall behavior, reinforcing the understanding of laminar-flow design trade-offs within the NACA 6-series.